

DIMITRIS OIKONOMOU

Charles Village, 21218, Baltimore, MD, USA | doikono1@jh.edu | <https://dimitris-oik.github.io/>

EDUCATION

Johns Hopkins University

PhD in Computer Science

- Advisor: Dr. Nicolas Loizou

Baltimore, USA

08/2023 - Present

National Technical University of Athens

MSc in Data Science and Machine Learning

- Thesis: Efficient Learning Algorithms of Time Varying Ranking Distributions, Advisor: Dr. Dimitris Fotakis

Athens, Greece

GPA: 9.8/10

University of Göttingen

MSc in Mathematics

- Thesis: Convergence Results for Entropic Transfer Operators, Advisor: Dr. Bernhard Schmitzer

Göttingen, Germany

GPA: 1.2/1 (~ 9.7/10)

National and Kapodistrian University of Athens

BSc in Mathematics

Athens, Greece

GPA: 9.6/10

RESEARCH

[7] **Dimitris Oikonomou**, Nicolas Loizou. *Curvature-Scaled Sharpness-Aware Minimization: Convergence Beyond Smoothness*. Under Submission.

[6] **Dimitris Oikonomou**, Matthew Buchholz, Yuen-Man Pun, Robert M. Gower, Nicolas Loizou. *Taking the Road Less Scheduled with Adaptive Polyak Steps*. In **NeurIPS 2026**

[5] **Dimitris Oikonomou** and Nicolas Loizou. *Adaptive Sharpness-Aware Minimization with a Polyak-type Step size: A Theory-Grounded Scheduler*. In **ICML 2026**

[4] **Dimitris Oikonomou** and Nicolas Loizou. *Safeguarded Stochastic Polyak Step-Sizes for Non-smooth Optimization: Robust Performance Without Small (Sub)Gradients*. In **ICML 2026**

[3] Robert Gower, Guillaume Garrigos, Nicolas Loizou, **Dimitris Oikonomou**, Konstantin Mishchenko, Fabian Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. In Transactions on Machine Learning Research, 2026

[2] **Dimitris Oikonomou** and Nicolas Loizou. *Sharpness-Aware Minimization: General Analysis and Improved Rates*. In **ICLR 2025**

[1] **Dimitris Oikonomou** and Nicolas Loizou. *Stochastic Polyak Step-sizes and Momentum: Convergence Guarantees and Practical Performance*. In **ICLR 2025**

HONORS & AWARDS

Mathematical Olympiads

- International Mathematical Olympiad (IMO): *Bronze Medal*
- Greek Mathematical Olympiad: *Silver Medal*
- Greek Mathematical Olympiad: *Bronze Medal*

Santa Marta, Colombia, 2013

Athens, Greece, 2014

Athens, Greece, 2013

Scholarships

- Hellenic Post Scholarship
- DAAD Scholarship
- JHU MINDS Fellowship

2015-2019

2019-2020

2024, 2025

TECHNICAL SKILLS

Programming Languages: Python, R, C/C++, MATLAB

Libraries: PyTorch, NumPy, scikit-learn, Pandas

Applications: OpenMP, CUDA, $\text{\LaTeX}$

TEACHING & PROFESSIONAL SERVICE

Teaching Assistant: *Computer Vision*

JHU, Fall 2024

Reviewer: IEEE Transactions on Signal Processing, TMLR, ICML, NeurIPS, ICLR, AISTATS

LANGUAGES

English (Fluent), **Greek** (Native), **French** (Basic)